

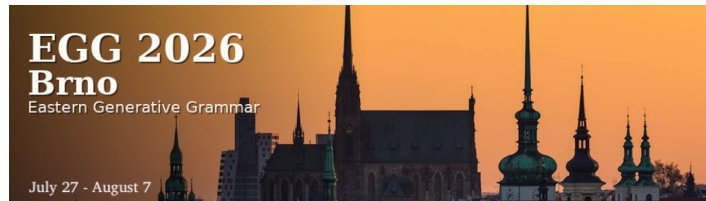


Minimalist Foundations

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<https://kxhoge.github.io/resources.html>



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Course outline

Monday	From X-bar theory to Bare Phrase Structure: The emergence of Merge
Tuesday	External and Internal Merge
Wednesday	Constraining Merge: Phase theory
Thursday	The Strong Minimalist Thesis, Merge, and the problem of labelling
Friday	The Miracle Creed: Can syntax be reduced to Merge?

External and Internal Merge

Outline

1. On Merge and Move
2. Agree
3. Displacement – from imperfection to virtual conceptional necessity
4. Summary and outlook

External and Internal Merge

On Merge and Move

- Chomsky (1995) introduces **Merge** as a syntactic operation.
- Merge forms a new syntactic object by combining two syntactic objects into an unordered set; the resulting object receives a label through the projection of one of the merged elements.

‘Applied to two objects α and β , Merge forms the new object K , eliminating α and β ... The simplest object constructed from α and β is the set $\{\alpha, \beta\}$, so we take K to involve at least this set, where α and β are the constituents of K . Does that suffice? Output conditions dictate otherwise; thus, verbal and nominal elements are interpreted differently at LF and behave differently in the phonological component. K must therefore at least (and we assume at most) be of the form $\{\gamma, \{\alpha, \beta\}\}$, where γ identifies the type to which K belongs, indicating its relevant properties. Call γ the label of K ’ (Chomsky 1995: 223).

‘...the label γ is either α or β ; one or the other projects and is the head of K . If α projects, then $K = \{\alpha, \{\alpha, \beta\}\}$ ’ (Chomsky 1995: 224).

External and Internal Merge

On Merge and Move

- Not all structure building involves Merge: in early Minimalism, **Move** remains a distinct operation responsible for creating displacement dependencies.

‘While Merge is costless for principled reasons, movement is not: the operation takes place only when forced (Last Resort); and it is overt, violating Procrastinate, only when that is required for convergence. If α has a strong feature F, it triggers an operation OP that checks F before the formation of a larger category that is not a projection of α . The operation OP may be Merge or Move’ (Chomsky 1995: 216).

External and Internal Merge

On Merge and Move

- In seeking to determine the extent to which a more computationally generic, domain-general conception of the faculty of language (FL) is viable, the Minimalist Program embraces both methodological and **substantive economy**. Grammars are therefore assumed to make minimal use of computational resources, with syntactic operations subject to a range of economy principles.

External and Internal Merge

On Merge and Move

- If α has a strong feature F, it triggers an operation (Merge or Move) that checks F before the formation of a larger category that is not a projection of α .
- Merge is free.
- Move is subject to **Last Resort**
(‘... a step in a derivation is legitimate only if it is necessary for convergence’; Chomsky 1995: 183; cf. Chomsky 1986).
- Move is subject to **Procrastinate**
(LF movement is “cheaper” than overt movement’; Chomsky 1995: 181).
- Move is subject to **Greed**.
(‘Last Resort ... is always “self-serving”: benefiting other elements is not allowed. Alongside Procrastinate, then, we have a principle of Greed : self-serving Last Resort’; Chomsky 1995: 184).

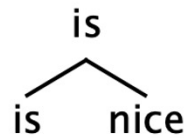
External and Internal Merge

On Merge and Move

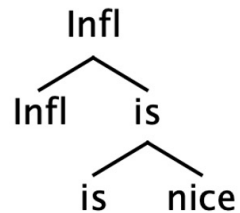
- (1) a. *John seems [that [___ is nice]].
b. John seems [___ to be nice].

(2) $N = \{(\text{John}, 1), (\text{seems}, 1), (\text{that}, 1), (\text{is}, 1), (\text{nice}, 1), (\text{Infl}, 2)\}$

(3) a.



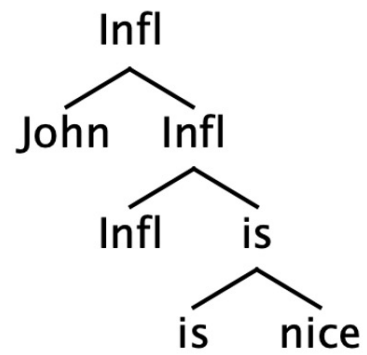
b.



External and Internal Merge

On Merge and Move

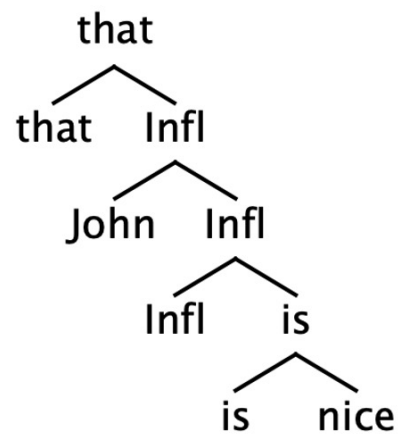
(3) c.



External and Internal Merge

On Merge and Move

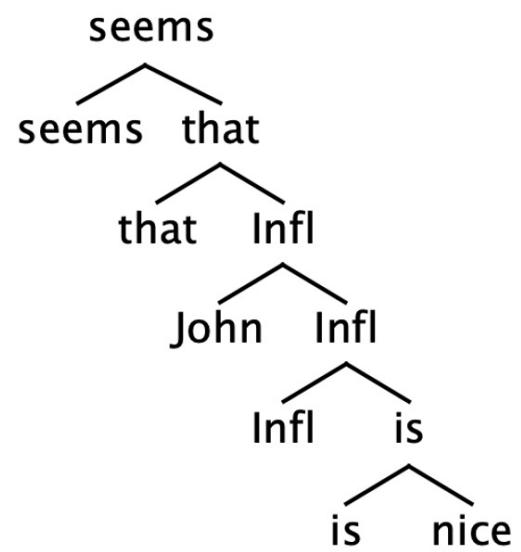
(3) d.



External and Internal Merge

On Merge and Move

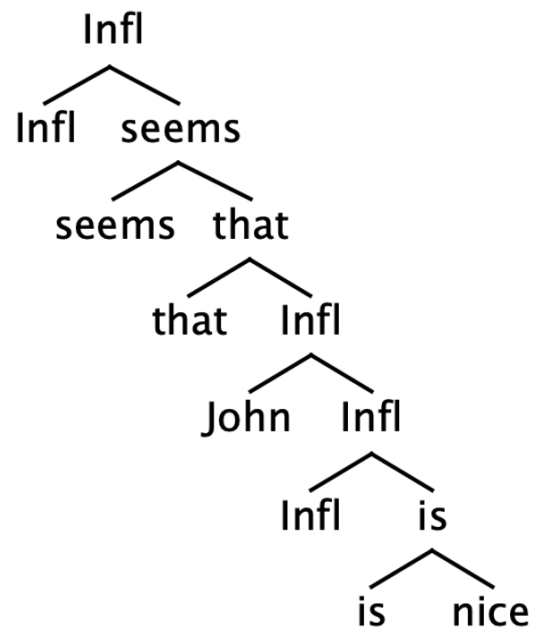
(3) e.



External and Internal Merge

On Merge and Move

(3) f.



External and Internal Merge

On Merge and Move

- In early Minimalism, lexical items enter the numeration with their syntactic features, including **Case**, already specified. Case features must be checked during the derivation. In the case of nominative Case, the Case feature of the nominal must be checked in a Spec–Head configuration with a (finite) inflectional head.
- Chomsky (1995) further distinguishes between **strong and weak features**: strong features must be checked before Spell-Out (the point in the derivation at which the syntactic representation is sent to the PF interface for pronunciation) for the derivation to converge.

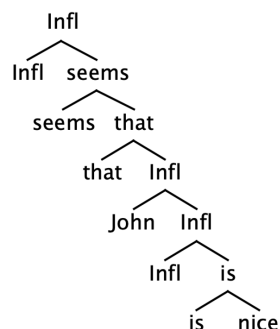
An example of a strong feature that requires checking before Spell-Out is the feature that derives the Extended Projection Principle (EPP), cf. ‘the Extended Projection Principle (EPP) plausibly reduces to a strong D-feature of I’ (Chomsky 1995: 213).

External and Internal Merge

On Merge and Move

- In **John seems that is nice*, *John* needs to check its nominative Case feature in a Spec-head configuration with Infl.
- Both embedded and matrix Infl bear a strong D-feature that requires checking against a matching D-feature in their respective specifiers.

(3f)



- *John* checks its nominative Case feature against the embedded Infl, and the embedded Infl checks its strong D-feature against *John*. Movement of *John* to the specifier of matrix Infl would be necessary for convergence, but, since it would benefit matrix Infl rather than *John*, movement of *John* would violate Greed.

External and Internal Merge

On Merge and Move

- A range of data appear to require a weakening of Greed (Collins 1997), including **successive-cyclic raising** constructions.

(4) John seems [____ to be likely [____ to be nice]].

- In (4), it is not clear what rules out *John* raising directly to the matrix SpecIP position without passing through the embedded SpecIP:

(5) John seems [to be likely [____ to be nice]].

Direct raising of *John* into the matrix clause would result in an EPP violation in the embedded clause, but, according to Greed, that should be irrelevant to the movement of *John*.

External and Internal Merge

On Merge and Move

- Chomsky (1995: 273) proposes to ‘reinterpret the operation of movement as “attraction”’, defining checking relations asymmetrically so that movement is driven by the checking requirements of the attracting head rather than by those of the moving constituent.
- (6) *Attract F*
K *attracts* F if F is the closest feature that can enter into a checking relation with a sublabel of K’
(Chomsky 1995: 273, ex. (84)).
- The introduction of Attract F paves the way for the Probe–Goal architecture of Chomsky (2000, 2001), shifting the initiation of syntactic dependencies from the moving constituent to the attracting head.

External and Internal Merge

Agree

- Checking theory predicts that agreement is established in a local Spec–Head configuration created by movement, but empirical evidence suggests that agreement need not always depend on displacement.

(7) a. **Strákarnir leiddust/*leiddist.**

[Icelandic]

the boys(Npl) walked-hand-in-hand(3pl/*3sg)

Strákunum leiddist/*leiddust.

b. **Strákunum leiddist/*leiddust.**

the boys(Dpl) bored(3sg/*3pl)

`The boys were bored.'

c. **Henni leiddust strákarnir.**

her(D) bored(3pl) the boys(Npl)

`She found the boys boring.'

(Sigurðsson 1996: 1, ex. (1), (2) & (3))

External and Internal Merge

Agree

- Chomsky (2000) dissociates agreement from syntactic movement by introducing the operation **Agree**, which establishes a feature-checking relation between a probing head and a matching goal. Move is now conceived as a combination of Merge and Agree.

First, what operations enter into this component of C_{HL} ? One is indispensable in some form for any language-like system: the operation *Merge*, which takes two syntactic objects (α , β) and forms $K(\alpha, \beta)$ from them. A second is an operation we can call *Agree*, which establishes a relation (agreement, Case checking) between an LI α and a feature F in some restricted search space (its *domain*). Unlike *Merge*, this operation is language-specific, never built into special-purpose symbolic systems and apparently without significant analogue elsewhere. We are therefore led to speculate that it relates to the design conditions for human language. A third operation is *Move*, combining *Merge* and *Agree*. The operation *Move* establishes agreement between α and F and merges $P(F)$ to αP , where $P(F)$ is a phrase determined by F (perhaps but not necessarily its maximal projection) and αP is a projection headed by α . $P(F)$ becomes the specifier (Spec) of α ([Spec, α]). Let us refer to *Move* of P to [Spec, ϕ] as *A-movement*, where ϕ is an agreement feature (ϕ -feature); other cases of *Move* are $\bar{A}$ -movement.

Plainly *Move* is more complex than its subcomponents *Merge* and *Agree*, or even the combination of the two, since it involves the extra step of determining $P(F)$ (generalized "pied-piping"). Good design conditions would lead us to expect that simpler operations are preferred to more

(Chomsky 2000: 101)

External and Internal Merge

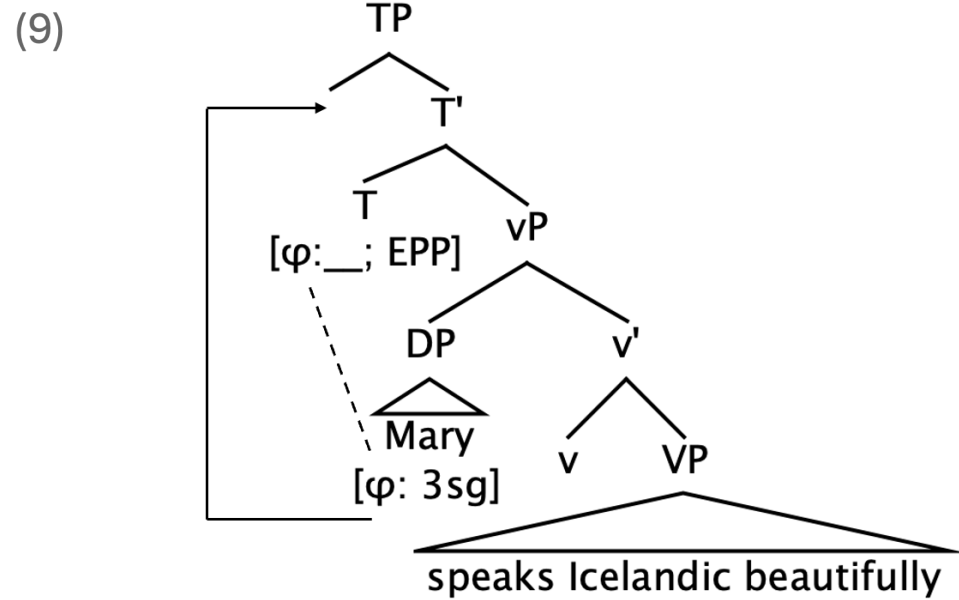
Agree

- Chomsky (2001) further develops the Probe–Goal architecture introduced in Chomsky (2000) by formalising **Agree as a feature valuation operation**. Functional heads bearing uninterpretable, unvalued features act as probes, searching their c-command domain for the closest active goal with matching interpretable, valued features.
- While Agree does not entail movement, movement is contingent on Agree. Chomsky (2001) suggests that movement is triggered by a so-called EPP-feature. A probing head bearing an EPP-feature attracts the agreed-with goal to its specifier.
- Case is valued as a by-product of Agree. ‘The value assigned depends on the probe: nominative for T, accusative for v’ (Chomsky 2001: 6).

External and Internal Merge

Agree

(8) Mary speaks Icelandic beautifully.



External and Internal Merge

Agree

- *Try your hand at some syntactic analysis*
- Consider once again the Icelandic example in (3) (repeated here for convenience):

(3) **Henni leiddust strákar.**

her(D) bored(3pl) the boys(Npl)

'She found the boys boring.'

Henni 'her' and *strákar* 'the boys' are the external and internal arguments, respectively, of *leiddust* 'bored'. Assume that *henni* is merged in SpecvP and *strákar* in the complement position of V. Draw the syntactic tree structure for the sentence, indicating the distribution of the features relevant to Agree and Move.

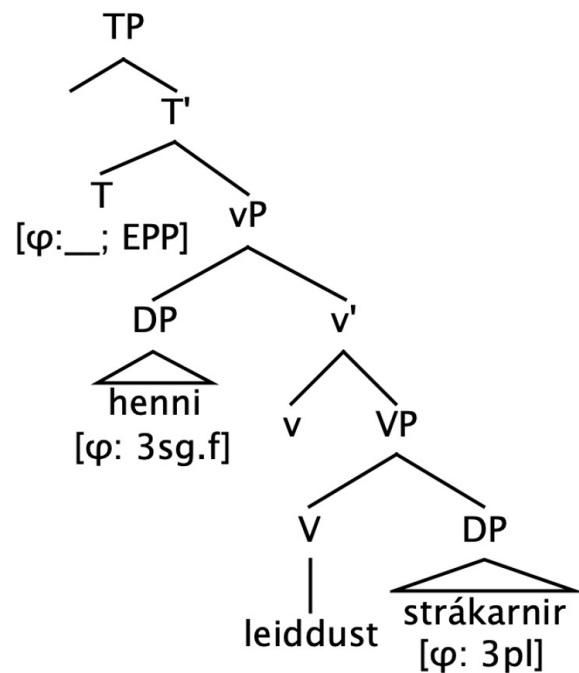
What problems, if any, do you encounter?

External and Internal Merge

Agree

- *Try your hand at some syntactic analysis*

(10)



Problem 1: The structurally closest DP (the dative experiencer) does not value the phi-features of T; instead, agreement is established with the lower nominative DP.

Problem 2: The DP that values T's phi-features is not the one that moves to SpecTP to satisfy the EPP, i.e. subject movement appears not to be contingent on Agree.

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External and Internal Merge

Displacement – from imperfection to virtual conceptual necessity

- In the Minimalist tradition, **displacement** (also called dislocation) refers to the phenomenon whereby a syntactic element is interpreted in one position but pronounced in another as the result of movement.
- (11) a. They elected an unpopular candidate.
b. An unpopular candidate was elected.
- If FL is organised as a maximally economical system that generates linguistic expressions legible at the sensorimotor and conceptual-intentional interfaces, C_{HL} should do no more than is necessary to satisfy the legibility conditions imposed by PF and LF. Displacement therefore appears to be an imperfection, introducing redundancy and additional computational complexity beyond what is required by the interfaces.

External and Internal Merge

Displacement – from imperfection to virtual conceptual necessity

The dislocation property (34b) is another apparent **imperfection**. In (35), for example, the phrase *an unpopular candidate* is in the natural position to be interpreted as object of *elect* in (35a,b) but not in (35c,d), though the interpretation is in relevant respects the same.

- (35) a. they [elected an unpopular candidate]
b. there was [elected an unpopular candidate]
c. an unpopular candidate was elected
d. there was an unpopular candidate elected

(Chomsky 2000: 119)

External and Internal Merge

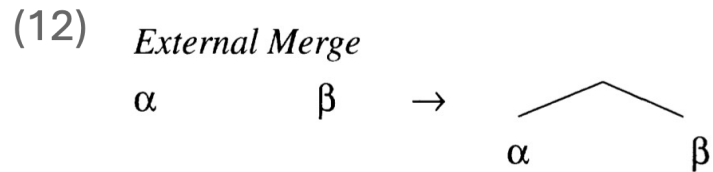
Displacement – from imperfection to virtual conceptual necessity

- While Chomsky (2001: 23) continues to analyse Move as a compound operation comprising Agree, pied-piping and Merge ('The compound operation Move consists of Agree, pied-piping, and Merge'), Chomsky (2004: 110) abandons Move as a syntactic operation, arguing instead that 'Merge is unconstrained, therefore either external or internal'.

External and Internal Merge

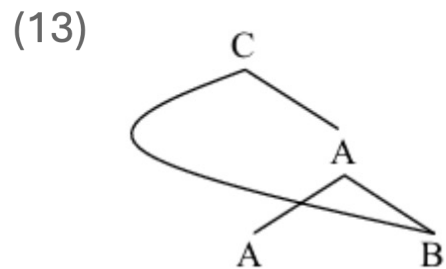
Displacement – from imperfection to virtual conceptual necessity

- **External Merge** combines two distinct syntactic objects to create a new syntactic object.



(Citko 2005: 475, ex. (1))

- **Internal Merge** combines a syntactic object with a subpart of that syntactic object, remerging the subpart at the root of the structure that already contains it.



(Citko & Gračanin-Yüksek 2020: 2, ex. (1b))

External and Internal Merge

Displacement – from imperfection to virtual conceptual necessity

- ‘Accordingly, displacement is not an “imperfection” of language; its absence would be an imperfection’ (Chomsky 2004: 110).
- ‘...there are two subcases of the operation Merge. Given A, we can merge B to it from outside A or from within A; these are external and internal Merge, the latter the operation called “Move”, which therefore also “comes free”, yielding the familiar displacement property of language. That property had long been regarded, by me in particular, as an “imperfection” of language that has to be somehow explained, but in fact it is a virtual conceptual necessity’ (Chomsky 2005: 12).

External and Internal Merge

Summary and outlook

- In its initial formulation (Chomsky 1995), the Minimalist Program introduced Merge as the fundamental operation for constructing hierarchical syntactic structure, retaining Move as a distinct operation responsible for deriving displacement. Over the following years, Move was progressively reconceptualised, first in terms of Attract and subsequently as a composite operation involving Agree, pied-piping, and Merge.
- With the recognition of External and Internal Merge (Chomsky 2004), syntactic movement in effect comes for free, leading to a new understanding of displacement as a virtual conceptual necessity rather than an imperfection.
- Chomsky (1995: 226) characterises Merge as ‘costless’, placing it outside ‘the domain of discussion of convergence and economy’. Once movement is reanalysed as Internal Merge, this raises the question of whether this property also holds true for Internal Merge: if Merge is costless, what constrains the remerging of a syntactic object already present in the derivation?

External and Internal Merge

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